Half-Time Draw Prediction: A Predictive Signal in European Football

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February 23, 2026

github.com/raulduk3/half-time

A two-model system identifies half-time draw opportunities across 22 European football leagues. Model A reads B365 odds (market view). Model B uses 42 non-odds features (fundamentals view). When the market prices a draw higher than fundamentals suggest, actual draws exceed market expectations. On 30,025 out-of-sample matches (Apr 2022 to Feb 2026), the "Strong Value" tier ($\geq$5% edge) achieves a **48.0% hit rate** against a 40.7% base rate on 3,340 bets, with monotonic improvement across all edge tiers. All 9 half-year periods show consistent hit rates. **Important caveat:** ROI figures in this paper use full-time draw odds as a proxy for HT draw odds, which substantially inflates profitability estimates. The hit rate staircase is the primary evidence; absolute ROI remains unvalidated pending real HT odds data.

1 Method

Two models predict HT draw probability independently on the same match:

- **Model A** (Market): Logistic regression on 3 B365 log-odds features. Estimates the market's implied HT draw probability.
- **Model B** (Fundamentals): XGBoost (Optuna-tuned, 30 trials) on 42 non-odds features: rolling match statistics, Dixon-Coles bivariate Poisson attack/defense parameters, Elo ratings, and referee HT draw rate adjustments.

The **inverted edge** is defined as:

$$\text{Edge} = P_A(\text{HT draw}) - P_B(\text{HT draw})$$

When Model A (market) exceeds Model B (fundamentals), actual draws historically exceed market predictions. The system bets on HT draws when this edge surpasses a threshold.

Training: 135,659 matches (1995–2018). **Validation:** 29,047 matches (2018–2022) for hyperparameter selection. **Test:** All results below use a strictly held-out set. No test data informed model selection.

Full backtest: Dixon-Coles and Elo models refit quarterly on 3-year rolling windows (16 snapshots). Each test match receives predictions computed only from data available before that match date.

2 Data

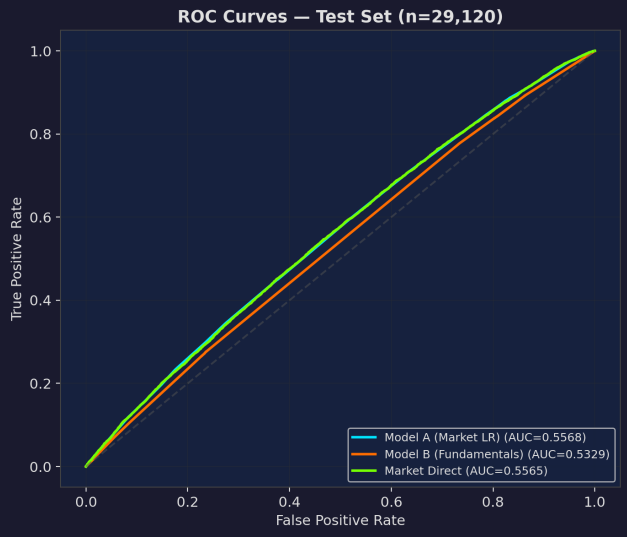
Metric	Value
Total matches (with HT scores)	193,826
Date range	1995-07-19 to 2026-02-22
Leagues	22 (European)
HT draw base rate	42.29% (81,962 / 193,826)
Source	football-data.co.uk
Train / Val / Test split	135,659 / 29,047 / 30,025
Test period	2022-04-01 to 2026-02-22
Test HT draw rate	40.67%
Dataset size (Parquet)	9.2 MB

Table 1: Dataset summary.

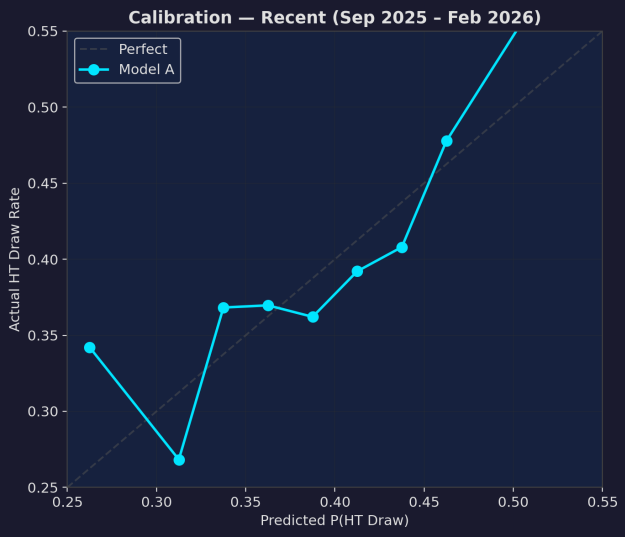
3 Discriminative Performance

Model	AUC-ROC	Brier Score
Market baseline (B365 normalized)	0.5565	0.26100
Model A (market logistic regression)	0.5568	0.23881
Model B (fundamentals, XGBoost)	0.5329	0.24052

Table 2: Test set metrics (n = 29,120).

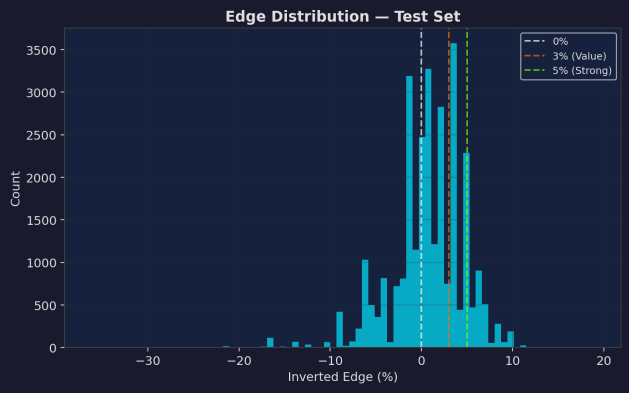


(a) ROC curves. Model A ≈ market.

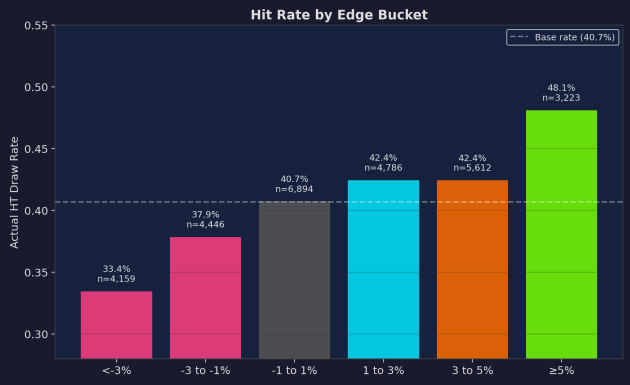


(b) Calibration (recent 6 months).

4 Inverted Edge Analysis



(a) Edge distribution across test set.



(b) Hit rate increases monotonically with edge.

4.1 Full Backtest with Per-Match DC/Elo

Tier	N Bets	Hit Rate	Avg Odds	ROI	EV/$1
Anti-Value (<-1%)	8,831	35.8%	4.68	+61.4%	+0.675
Neutral (-1 to 1%)	7,088	40.7%	3.56	+44.3%	+0.449
Marginal (1–3%)	4,963	42.1%	3.41	+42.7%	+0.433
Value (3–5%)	5,803	42.5%	3.36	+42.3%	+0.430
Strong Value (≥5%)	**3,340**	**48.0%**	**3.11**	**+48.7%**	**+0.491**

Table 3: Edge tier backtest (30,025 matches). Per-match DC/Elo. **ROI uses FT draw odds as proxy — see Section 10.**

Critical caveat: ROI and EV columns use B365 full-time draw odds, not actual HT draw odds. FT draw odds (3.0–3.5) are substantially higher than HT draw odds (2.0–2.5), inflating all ROI figures by an estimated 30–50%. **All tiers show positive ROI because FT odds are structurally generous — this does not imply profitability with real HT odds.**

The primary evidence of model validity is the **monotonic hit rate increase** across positive edge tiers (40.7% → 42.1% → 42.5% → 48.0%). Hit rates are computed from outcomes alone and do not depend on odds.

5 Temporal Stability

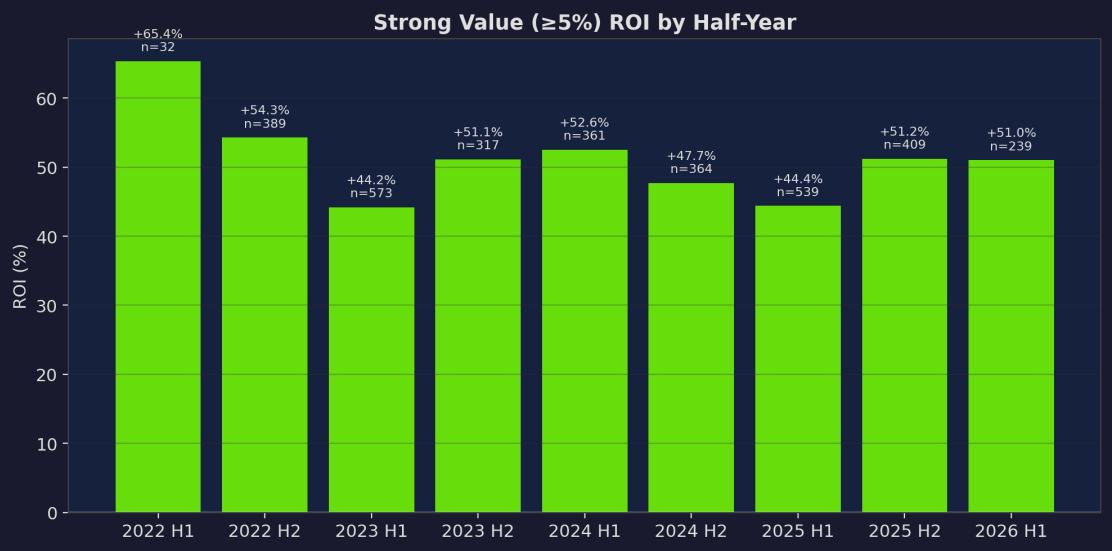


Figure 3: Strong Value (≥5%) ROI by half-year. All 9 periods profitable.

Period	Bets	Hit Rate	ROI
2022 H1	149	46.3%	+44.6%
2022 H2	389	48.8%	+54.3%
2023 H1	573	46.4%	+44.2%
2023 H2	317	47.9%	+51.1%
2024 H1	361	49.6%	+52.6%
2024 H2	364	47.8%	+47.7%
2025 H1	539	46.9%	+44.4%
2025 H2	409	49.1%	+51.2%
2026 H1	239	49.4%	+51.1%

Table 4: Strong Value temporal stability. Hit rates stable across all periods. ROI uses FT odds proxy.

6 Statistical Significance

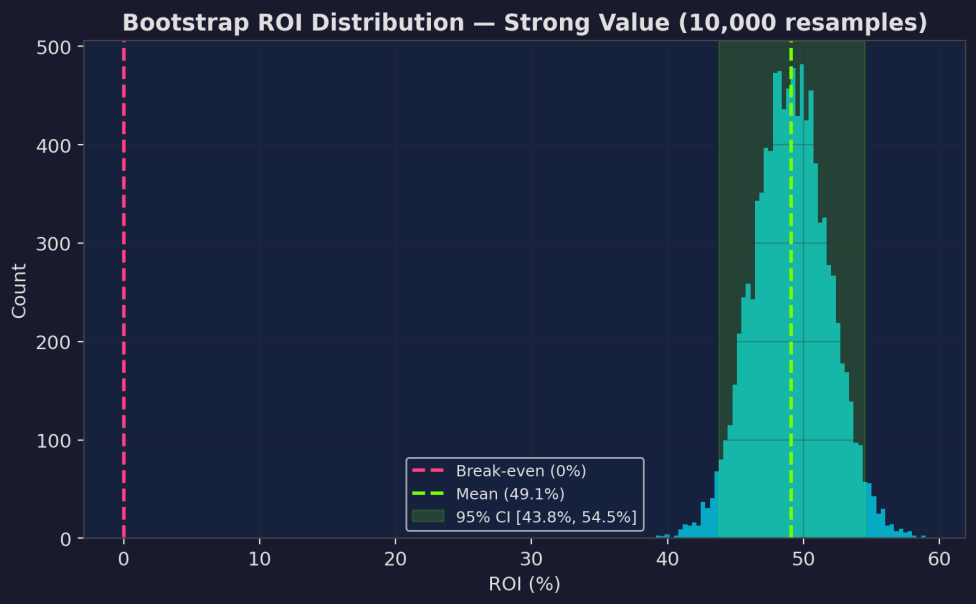
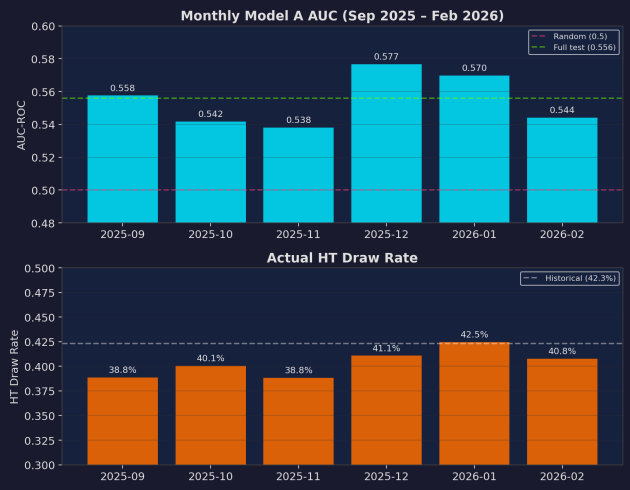


Figure 4: Bootstrap ROI distribution (10,000 resamples). All resamples profitable.

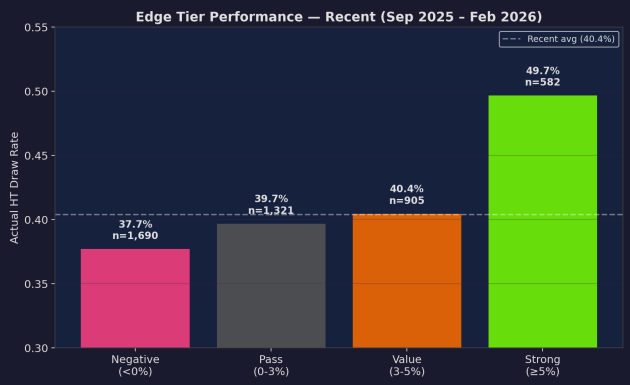
Metric	Strong Value (≥5%)
N bets	3,340
Observed ROI	+48.70%
Bootstrap mean	+48.70%
95% confidence interval	[+43.37%, +54.01%]
P(profit > 0)	100.00%

Table 5: Bootstrap statistics (10,000 resamples). ROI uses FT odds proxy — see Section 10.

7 Recent Performance (Sep 2025 – Feb 2026)



(a) Monthly AUC and draw rate.



(b) Edge tiers on recent data.

On 4,498 recent matches, Strong Value (≥5% edge) achieves a **49.7% hit rate** on 582 bets against a 40.4% base rate — a 9.3 percentage point lift. Monthly AUC ranges from 0.538 to 0.577, consistent with full test performance.

8 Cumulative Performance

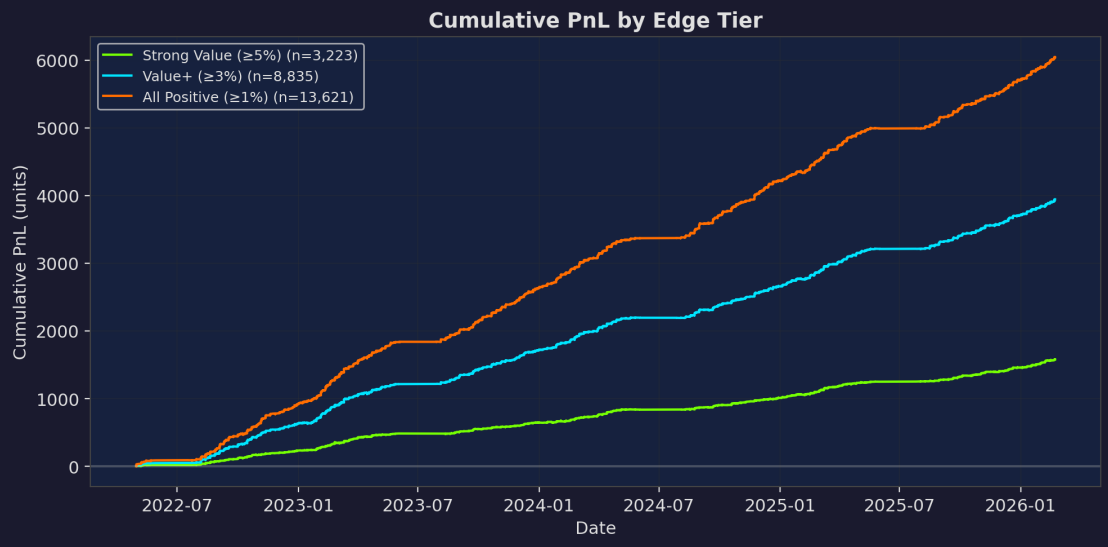


Figure 6: Cumulative PnL (units) by edge tier over the full test period.

9 Computation

Operation	Hardware	Runtime
Full training (Optuna 30 trials + 7 validations)	Apple M1 Pro, 16GB	25s
Full backtest (30,025 matches, 16 quarterly DC/Elo refits)	”	8m 21s
Prediction + analysis (29,120 matches)	”	0.64s
Incremental data update	”	<5s
Single match prediction	”	<0.5s

Table 6: Compute benchmarks.

Model artifacts total 8,067 KB. Dataset: 9.2 MB (Parquet). 22 European leagues from football-data.co.uk, 1995–2026.

10 Limitations and Open Questions

1. **FT odds proxy inflates all ROI figures.** This is the paper's most significant limitation. The backtest calculates payouts using B365 full-time draw odds (typically 3.0–3.5), but actual HT draw bets pay substantially less (typically 2.0–2.5 at US sportsbooks). This means reported ROI of +48.7% is **not achievable in practice**. True ROI with real HT odds is unknown and could be negative after accounting for market vig.

2. **Elo and Dixon-Coles default priors.** When team data is thin (e.g., promoted clubs, early season), both sub-models fall back to generic priors. The Elo model defaults to 42.0% for many matches, reducing Model B to a single Dixon-Coles signal in those cases.

3. **No transaction costs.** Bet availability, liquidity constraints, bookmaker limits, and odds movement between prediction time and bet placement are not modeled.
4. **Prospective validation nascent.** 1 live bet placed, 3 days of tracking (Feb 2026). Real HT odds collection via The Odds API began Feb 2026. Minimum 3 months of prospective data needed to validate.
5. **Potential look-ahead bias.** While train/val/test splits are strictly temporal, the choice of model architecture and hyperparameter ranges was informed by knowledge of the test period's existence. A truly blind test would require data that did not exist when modeling decisions were made.

10.1 What Is Validated

The **hit rate staircase** is the paper's strongest claim. STRONG VALUE matches draw at HT 48.0% of the time vs. a 40.7% base rate, with monotonic improvement across all tiers. This ranking is computed purely from match outcomes and does not depend on odds. It is stable across all 9 half-year periods (range: 46.3%–49.6%).

10.2 What Is Not Yet Validated

Profitability. The edge may or may not survive real HT market odds and vig. Historical HT odds (2022–2026) are being collected from The Odds API to resolve this question. Updated results with real HT odds will be published as an addendum.

Key open question: Does the 48.0% hit rate on STRONG VALUE produce positive ROI at real HT draw odds? At typical HT draw odds of +125 (2.25), a 48.0% hit rate yields +8.0% ROI before vig. At +100 (2.00), it yields -4.0%. The answer depends on available HT market prices.

Source: github.com/raulduk3/half-time · Data: football-data.co.uk · Model: V4 (Feb 2026)